Circles Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Radius vs. diameter

ORIGINAL PROBLEM

Find the area of a circle with diameter 10.

INCORRECT WORKED SOLUTION

$A = \pi r^2$

$A = \pi(10)^2$

$A = 100\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Circumference

ORIGINAL PROBLEM

Find the circumference of a circle with diameter 12.

INCORRECT WORKED SOLUTION

$C = 2\pi r$

$C = 2\pi(12)$

$C = 24\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Circles Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

3. Arc length

ORIGINAL PROBLEM

Find the length of an arc with central angle 60° in a circle of radius 9.

INCORRECT WORKED SOLUTION

$$\text{Arc} = \frac{60}{360} \cdot \pi r^2$$

$$\text{Arc} = \frac{60}{360} \cdot \pi(81)$$

$$\text{Arc} = \frac{27\pi}{2}$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Sector area

ORIGINAL PROBLEM

Find the area of a sector with central angle 90° and radius 4.

INCORRECT WORKED SOLUTION

$$\text{Area} = \frac{90}{360} \cdot 2\pi r$$

$$\text{Area} = \frac{1}{4} \cdot 2\pi(4)$$

$$\text{Area} = 2\pi$$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Circles Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

5. Inscribed angle

ORIGINAL PROBLEM

An inscribed angle intercepts an arc of 80°. Find the measure of the inscribed angle.

INCORRECT WORKED SOLUTION

Inscribed angle = arc

Inscribed angle = 80°

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Angle inside a circle

ORIGINAL PROBLEM

Two chords intersect inside a circle. The arcs intercepted by the angle and by its vertical angle measure 50° and 90°. Find the angle.

INCORRECT WORKED SOLUTION

$$\text{Angle} = \frac{90 - 50}{2}$$

Angle = 20°

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Circles Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

7. Tangent lines

ORIGINAL PROBLEM

Tangent segment PT touches a circle at T. The radius OT = 5 and OP = 13. Find PT.

INCORRECT WORKED SOLUTION

PT = OP − OT

PT = 13 − 5 = 8

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. Intersecting chords

ORIGINAL PROBLEM

Two chords intersect inside a circle. One is cut into segments 4 and x. The other is cut into 6 and 8. Find x.

INCORRECT WORKED SOLUTION

4 + x = 6 + 8

x = 10

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Circles Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

9. Two tangents

ORIGINAL PROBLEM

Two tangent segments from the same point P have lengths PA = 3x + 2 and PB = 5x − 10. Find x.

INCORRECT WORKED SOLUTION

PA + PB = 90

3x + 2 + 5x − 10 = 90

8x = 98

x = 12.25

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Angle in a semicircle

ORIGINAL PROBLEM

AB is a diameter and C is a point on the circle. Find m∠ACB.

INCORRECT WORKED SOLUTION

Arc AB is a semicircle: 180°.

m∠ACB = 180°

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points